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"SQE Assignment 1"

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S24-BSE-013

- CMMI levels -
- Why and how IT companies follow it

CMMI Level 1: Initial

- Process Areas (0):
There are no process areas at level 1. Work is ad hoc, reactive, and driven by individual effort and crisis rather than documented procedures. Success is rarely repeatable.
- Why and how IT companies follow it:
  - Developed Countries: Tech startups, innovation hubs, and early R and D teams intentionally stay at level 2. Their priority is extreme speed, rapid iteration and building MVPs without being slowed down by documentation/approval gates.
  - Developing Countries: Small local software houses, boutique agencies, and freelance shops operate here mostly by default. They face tight budget constraints, lack formal quality training, and work with clients who do not mandate formal Quality Standards.

CMMI Level 2: Managed

- Process Areas (7):
  1. CM (Configuration Management): Version control

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and baselining of code and project deliverables.

2. MA (Measurement and Analysis): collecting and analysis of metrics like effort, defects and schedule variance.

3. PMC (Project Monitoring and Control): Tracking ongoing work against plans and taking corrective action when milestones slip.

4. PP (Project Planning): Developing realistic project estimates, schedules, budgets and source plans.

5. PPQA (Process and Product Quality Assurance): Auditing processes and deliverables to ensure standards are met.

6. REQM (Requirements Management): Tracking client requirements and ensuring changes are consistently mapped to code and tests.

7. SAM (Supplier Agreement Management): Managing external vendors, third-party libraries, and outsourced subcontractors.

- Why and how IT companies follow it:
  - Goal shift from personal heroics to project-level discipline and repeatability
  - Developed countries: Mid-sized IT firms and corporate engineering divisions adopt level 3 practices (often through scrum/kanban setups) to setup budget overruns, prevent release chaos, and maintain solid governance over external SaaS/cloud vendors.

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- Developing Countries: Offshore development centers use level 2 as an entry ticket to international market. It proves to western clients that deliveries are dependable even if individual engineers leave, and matches like RFAM protect firms from absorbing the cost of unpaid scope creep.

CMMI level 3 : Defined

- Process areas (11) :

1- DAR (Decision Analysis and Resolution): Evaluating formal trade-offs using weighted scoring for major technical decisions.

2. IPM (Integrated Project Management) Managing projects using tailored versions of the company's standard process framework.

3. OPD (Organizational Process definition) Maintaining organizational process assets, lifecycles models, templates and guidelines.

4. OPF (Organizational Process Focus): Continuously identifying, planning and deploying organizational process improvements

5. OT (Organizational training) Delivering structured training so staff can execute standardized process effectively.

6. PI (Product Integration) Step-by-step assembly and interface testing of system modules and third party components.

7. RD (Requirement Development): Eliciting, analysing and validating customer and product requirements

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before development starts

8. RSKM (Risk Management) Proactively identifying, scoring, and mitigating technical and project risks before they cause failures.

9. TS (Technical Solution) Designing, architecting, and implementing robust code and software architectures

10. VAL (Validation) Demonstrating the final system satisfies user needs in real-world conditions ("building the right product").

11. VER (Verification) Ensuring each artifact correctly satisfies the specifications established at each phase ("building the product right").

- Why and how IT companies follow it.

  - Goal: Elevate discipline from individual isolated projects to an organization-wide standard set of processes (OSSP)

  - Developed Countries: Essential for prime contractors bidding on government, defense, aerospace, healthcare, and critical infrastructure contracts where formal CMMI level 3 appraisals are legally required. Large enterprises also use it to maintain architectural consistency across global offices.

  - Developing countries: Tier-1 IT exporters and large tech services companies adopt level 3 to win lucrative global enterprise RFP's and multi-year maintenance contracts. Crucially, organizational process assets (OPD) and structured training (OT) protect firms against knowledge loss caused by high industry employee turnover.